Historical Maps of the Renaissance & Early Modern World

An Instructional Cartographic Guide for ArcGIS Story Maps

Seven Historical Maps — Volume II

Maps Covered: Map of Imola (da Vinci, 1502) • Universalior Cogniti Orbis Tabula (Ruysch, 1507) • Map of Utopia (A. Holbein, 1518) • Carta Marina (Olaus Magnus, 1539) • Septentrionalium Terrarum (Mercator/Hondius, 1606) • Nautical Chart/Mao Kun Map (Zheng He, pub. 1628) • Hereford Mappa Mundi (c. 1300)

Map 1: Map of Imola — 1502 (Leonardo da Vinci)

The Cartographer

Leonardo da Vinci (1452–1519) was the quintessential Renaissance polymath — painter, sculptor, architect, anatomist, engineer, and scientist — whose insatiable curiosity led him into virtually every field of human inquiry, including cartography. Born in Vinci, near Florence, he rose to fame under the patronage of Ludovico Sforza of Milan and later served as chief military architect and engineer to the notorious Cesare Borgia, in which capacity he created the masterpiece of scientific mapmaking known as the Plan of Imola.

Historical Facts About the Map

- In August 1502, Leonardo was appointed General Architect and Engineer to Cesare Borgia, the ambitious son of Pope Alexander VI, who had seized the city of Imola three years earlier. Borgia needed a detailed, accurate plan of his new city for military strategy and administrative control, and tasked the 50-year-old Leonardo with creating it. The map was produced while Leonardo was simultaneously working on other projects — it was made between his completion of The Vitruvian Man and the painting of the Mona Lisa.
- Leonardo physically walked every street of Imola, taking measurements with a hodometer (a wheel-based distance-measuring device he had recently developed) and taking compass bearings from the tower of the Palazzo Comunale at the city's central crossroads. The map was then laid out using polar coordinates — a geometric technique proposed by Florentine humanist Leon Battista Alberti — producing a true overhead or "ichnographic" view, one of the first of its kind in European cartography.
- The map is remarkable for its accuracy: historians have confirmed that it can still be used to navigate the streets of Imola today, more than 520 years after it was drawn. This precision was revolutionary — contemporary maps of cities used oblique, artistic perspectives that were decorative but useless for navigation.
- Drawn with chalk, ink, and wash on paper, the map measures approximately 44 x 60 cm. It is divided into eight compass sectors, each further subdivided, and in the margins Leonardo recorded distances and directions to neighboring towns in his characteristic mirror writing. The original is held in the Royal Collection at Windsor Castle (RCIN 912284).
- The Imola Plan is considered the world's oldest surviving ichnographic city map — a true "bird's-eye view" plan where all features are rendered to accurate scale from directly overhead — which is the standard format used in all city planning and architectural mapping today. This single drawing effectively invented modern city cartography.

Points of Interest — Scavenger Hunt Style

POI #1 — The Eight Compass Points on the Circular Border

Look for… Around the outer ring of the map, look for eight evenly spaced points where four straight lines cross the circular boundary of the city. At each point, Leonardo has written the name of a wind direction in his own handwriting.

This compass ring was Leonardo's revolutionary contribution to city mapping — rather than orienting buildings pictorially (as all previous city views had done), he aligned the entire plan to true compass bearings. This meant the map could be physically used on the ground, with a compass, to navigate any street in Imola. It was the precursor of every city map ever printed since.

POI #2 — The Rocca Sforzesca Fortress

Look for… On the left (western) side of the circular city plan, look for a large square or rectangular structure that projects into the moat — a prominent fortified complex with thick walls set apart from the urban street grid.

The Rocca Sforzesca was the military stronghold of Imola, originally built by Galeazzo Sforza in the 1470s and later the residence of the famous Countess Caterina Sforza, whom Cesare Borgia had recently defeated and captured. For Borgia, understanding the layout of this fortress was a core purpose of commissioning the map — military control of the Rocca meant military control of the city.

POI #3 — The City Moat and Circular Wall

Look for… Trace the clear circular boundary of the city itself — a defensive ring wall surrounded by a moat or water channel, separating the urban core from the agricultural land and approach roads visible beyond.

Imola was a heavily fortified medieval city, and its circular wall system was a classic example of late medieval defensive architecture. Cesare Borgia's plans for Imola included improvements to these fortifications, and Leonardo was specifically empowered to order military improvements as well as to survey. The moat system was one of the features Leonardo documented in detail.

POI #4 — The Santerno River at the Bottom

Look for… At the bottom edge of the circular map, look for a sinuous, free-flowing curve representing the Santerno River — noticeably more fluid and organic than the sharp geometric lines of the streets above it.

In a map otherwise defined by mathematical precision and straight angles, Leonardo allowed himself one artistic indulgence: the river is drawn as a naturally flowing form, reflecting his deep observation of water in motion. This detail reveals the dual nature of his genius — the scientist who measured every street also remained the artist who saw the river as it truly moved.

POI #5 — The Street Grid and Block Structure

Look for… Look at the interior of the city for the irregular street grid — a network of lanes and blocks that does not follow a regular Roman grid but meanders organically, reflecting centuries of medieval urban growth.

The irregularities in the street plan were deliberately preserved by Leonardo because he was recording reality, not idealism. He noted that some building facades were slightly misaligned, details so small as to seem irrelevant — but Leonardo's mandate was accuracy above all. This fidelity to observed fact rather than idealized convention was the philosophical breakthrough that made the Imola plan a turning point in Western cartography.

POI #6 — The Palazzo Comunale Tower at the Center

Look for… Near the center of the map, find the main civic building — the Palazzo Comunale (Town Hall) — identifiable as one of the prominent labeled structures at or near the central crossroads of the city.

Leonardo took his primary compass bearings from the tower of the Palazzo Comunale, using it as the geometric anchor point for the entire plan. Everything else was positioned relative to this central landmark. This method — establishing a fixed reference point and mapping outward by compass and measured distance — was a mathematical innovation that anticipated the triangulation techniques used in modern surveying.

Web Links

Digital Viewing (Royal Collection Trust, Windsor — high resolution zoom):
https://www.rct.uk/collection/912284/a-map-of-imola

Print-Optimized Reference (National Geographic article with full image):
https://www.nationalgeographic.com/history/history-magazine/article/leonardo-da-vinci-transformed-mapping-art-science

Map 2: Universalior Cogniti Orbis Tabula — 1507 (Johannes Ruysch)

The Cartographer

Johannes Ruysch (c. 1460–1533) was a Flemish-born explorer, Benedictine monk, astronomer, manuscript illuminator, painter, and cartographer from Utrecht who led one of the most remarkable multi-disciplinary lives of the early Renaissance. He is believed to have participated in at least one transatlantic voyage to the New World in the late 1490s — possibly accompanying John Cabot's 1497 expedition to Newfoundland — before eventually settling in Rome, where he befriended Raphael and was commissioned by Pope Julius II to create his groundbreaking world map of 1507.

Historical Facts About the Map

- The Ruysch Map (full title: Universalior Cogniti Orbis Tabula Ex Recentibus Confecta Observationibus, meaning "Universal Map of the Known World Prepared from Recent Observations") was published in Rome in 1507–1508 as part of a new edition of Ptolemy's Geographia. It is the second oldest known printed map to depict the New World, preceded only by the Contarini-Rosselli map of 1506 (which survives in only a single copy). The Ruysch map, by contrast, was printed in multiple copies and thus widely distributed — scholars have identified 64 surviving examples — making it the most broadly influential early printed depiction of the Americas.
- The map was engraved on a distinctive fan-shaped coniform (equidistant conic) projection, placing the Atlantic Ocean at its center rather than at the edge — a radical departure from Ptolemaic tradition. This projection choice reflected Ruysch's own practical experience as a navigator: he knew that the Atlantic was the crucial space connecting the known world to the new discoveries.
- In accordance with Columbus's own convictions, Ruysch depicted North America (labeled Terra Nova) as a protrusion of the Asian continent rather than a separate landmass. South America appears as a large separate body labeled Mundus Novus (adopting Amerigo Vespucci's term), while Hispaniola (Spagnola) floats ambiguously between the two. This confusion captures cartographers' honest bewilderment in real time — the world was changing faster than knowledge could keep up.
- The map featured the first serious printed attempt to depict the polar regions, showing four large landmasses arranged around the North Pole — drawn from the legendary Inventio Fortunatae, a now-lost medieval account of Arctic exploration. These phantom polar lands would influence maps of the Arctic for more than a century.
- The Ruysch Map drew heavily on Portuguese geographical sources, reflecting Ruysch's later service as astronomer and cartographer to King Manuel I of Portugal. It features notably accurate depictions of the African and South Asian coastlines derived from Portuguese voyages, including a reference to Portuguese contact with Sri Lanka in 1505 — making it one of the most up-to-date maps of the Indian Ocean in existence at the time of its publication.

Points of Interest — Scavenger Hunt Style

POI #1 — Mundus Novus — South America

Look for… In the lower left (southwestern) area of the fan-shaped map, find a large landmass floating in open ocean, labeled with the Latin words "Mundus Novus" (New World) or a variant spelling. Look for Portuguese place names scattered along its northern coast.

This is one of the earliest printed depictions of South America. The name Mundus Novus was coined by Amerigo Vespucci himself, and Ruysch adopted it here to acknowledge that this was indeed a previously unknown continent — not merely islands off the coast of Asia. The over 25 Portuguese place names along the coast reflect the source of Ruysch's geographic intelligence: direct access to Portuguese voyage records.

POI #2 — Terra Nova — North America Attached to Asia

Look for… In the upper left portion of the map, find a landmass labeled "Terra Nova" that extends leftward and appears to merge with or connect to the large Asian landmass on the right side of the map. Notice the mountains labeled across this zone.

This attachment of North America to Asia captures Columbus's dying conviction that he had reached Asia, not a new continent. Ruysch, having likely sailed to Newfoundland himself with Cabot, is here choosing Columbus's model over Vespucci's. This cartographic debate — Asia or new world? — was the defining intellectual question of the age, and this map presents one side of the argument in print.

POI #3 — Spagnola and the Caribbean

Look for… Between the two large landmasses, look for a cluster of smaller islands. The largest is labeled "Spagnola" (Hispaniola), and nearby should be another large mass that Ruysch tentatively identifies with Marco Polo's Cipangu (Japan).

Ruysch openly admits in a scroll text on the map that he is uncertain whether "Spagnola" is Japan or a new island — a remarkably honest acknowledgment of cartographic uncertainty in an era when most mapmakers disguised their ignorance. The uncertainty itself is historically valuable: it shows that European explorers in 1507 were still genuinely confused about what they had found.

POI #4 — Four Polar Islands Around the North Pole

Look for… At the very top of the fan-shaped map (the far north), look for four large landmasses arranged symmetrically around a central polar zone, labeled in Latin. Between them are channels of water flowing toward a whirlpool at the center.

Ruysch based this Arctic design on the Inventio Fortunatae, a medieval English account (now lost) of a journey to the far north. The four polar islands and the central whirlpool would be repeated by Mercator and many others throughout the 1500s. These phantom lands illustrate how ancient texts and legends shaped cartographic "knowledge" long after they could have been verified or disproven.

POI #5 — The Fan-Shaped Coniform Projection

Look for… Step back and observe the overall shape of the map: rather than a rectangle or oval, the whole world is arranged in a fan or sector shape, wider at the top (north) than at the bottom, like a slice of a cone.

Ruysch's choice of a conic projection was sophisticated and mathematically informed. He was one of the few cartographers in 1507 who understood that transposing a globe onto a flat surface required a systematic geometric approach. The conic projection minimized distortion

across the mid-latitudes — the most actively navigated zones — making this a more practical map for actual use than most of its contemporaries.

POI #6 — The African Cape and Indian Ocean

Look for… On the right-hand side of the map, find the unmistakable triangular southern tip of Africa, with the Indian Ocean opening to the east. Look for Portuguese place names along the east African coast and across to India.

Ruysch's African and Indian Ocean coastlines were among the most accurate in print at the time of publication, drawn from Portuguese sources unavailable to most other European cartographers. Portugal's successful rounding of the Cape of Good Hope by Vasco da Gama in 1498 had transformed the Indian Ocean from a cartographic mystery into a known navigational space, and Ruysch was among the first to capture this knowledge in a printed world map.

Web Links

Digital Viewing (Library of Congress, high resolution): https://www.loc.gov/item/2003626426/

Print-Optimized Reference (Stanford Leventhal Map Collection): https://collections.leventhalmap.org/search/commonwealth:3f462s18s

Map 3: Map of Utopia — 1518 (Ambrosius Holbein)

The Cartographer

Ambrosius Holbein (c. 1494–c. 1519) was a Swiss-German painter, draughtsman, and woodcut artist from Augsburg, and the older brother of the more celebrated Hans Holbein the Younger. Though he died young — probably in his mid-twenties — Ambrosius was a highly skilled and thoughtful artist who worked primarily in Basel, producing portraits, book illustrations, and decorative woodcuts for leading Renaissance printers including Johann Froben. He is remembered today primarily for this single extraordinary cartographic woodcut, the Map of Utopia for Sir Thomas More's 1518 Froben edition, which stands as one of the most philosophically complex images in the history of printed maps.

Historical Facts About the Map

- Sir Thomas More first published Utopia in Antwerp in 1516, but the most significant illustrated edition was the Basel edition of 1518, printed by Johann Froben and overseen by the great humanist scholar Erasmus of Rotterdam, More's closest friend. It was Erasmus who likely commissioned the woodcut map from Ambrosius Holbein, making this one of the most intellectually charged map commissions in history — Erasmus understood that a visual representation of Utopia would give More's abstract ideal society a concrete, spatial reality.
- The name "Utopia" was deliberately coined by Thomas More from Greek: ou-topos means "no place," while eutopia means "good place." The double meaning — simultaneously "nowhere" and "the ideal" — was the central philosophical joke of the book. The map makes this pun physically real: it depicts a detailed, plausible-looking island that the reader's eye wants to believe in, while the joke labels (Amaurotum = "Mist-town," Anydrus = "Waterless river") announce at every turn that this is a fiction.
- Art historians have identified a hidden skull within the overall composition of the map, concealed within the shapes of the crescent-shaped island and the sailing ship at its base — the ship forms the jaw and teeth, the island the skull. This memento mori (reminder of death) embedded within a utopian vision was entirely characteristic of Renaissance art's unsettling duality: the perfect society is also the land of the dead, "no place" because it cannot exist while human mortality persists.
- The map shows the island of Utopia as a crescent-shaped landmass — More described it as having been cut from the mainland by its founder, King Utopus, who dug a channel 15 miles wide across the isthmus. The island contains 54 cities, all identical in plan, reflecting More's vision of a society without inequality, private property, or social hierarchy.
- The Map of Utopia inaugurated the tradition of literary cartography — the mapping of fictional worlds — that would extend through Jonathan Swift's Gulliver's Travels, Robert Louis Stevenson's Treasure Island, J.R.R. Tolkien's Middle-earth, and countless other imaginary geographies. Every fictional map ever drawn owes something to this small woodcut.

Points of Interest — Scavenger Hunt Style

POI #1 — The Crescent Shape of the Island

Look for… Look at the overall form of the island itself — a curved, crescent or horseshoe shape that faces inward toward a protected central harbor. Several small towns dot the coastline.

More described in his text that the island was originally a peninsula until King Utopus ordered his conquered people to cut a 15-mile channel, separating it from the mainland. The crescent shape creates a natural harbor — More's Utopia is designed for trade and maritime contact with the outside world, not isolation. The shape is both defensive and economically rational: typical of the humanist thinking that imagined ideal societies as ordered, functional, and rational rather than merely pious.

POI #2 — Amaurotum — The Capital City

Look for… Find the label for Amaurotum on the map, positioned as the central or most prominent city on the island. The name comes from the Greek amaurotos, meaning "obscure" or "shadowy."

The capital city's name is More's most pointed joke: the capital of the ideal state is literally called "Mist-town" — it exists in a haze, visible from a distance but never quite reached. This is More at his most subtle: the utopian capital is the most desirable destination imaginable, and its name tells you precisely that you will never get there. For Renaissance humanists trained in Greek, the irony would have been immediately visible.

POI #3 — The River Anydrus

Look for… Find the river flowing through or around the island, labeled "Anydrus." The name derives from the Greek an-hydros, meaning "without water."

The "Waterless River" is another of More's delightful impossibilities, embedded directly into the cartographic label. The river is drawn clearly — a river without water, mapped with perfect cartographic authority. This technique of giving plausible geographic form to logical absurdity was More's strategy throughout: Utopia looks real precisely because the lies are told in the language of fact.

POI #4 — Raphael Hythloday at Lower Left

Look for… In the lower left corner of the woodcut, find a small figure in Renaissance dress standing on a shore, gesturing toward the island. This is Raphael Hythloday, the fictional explorer-traveler who discovered Utopia.

Hythloday's name is itself a Greek pun: hythlos means "nonsense" and daios means "skilled." He is the "skilled purveyor of nonsense" — the fictional narrator who makes Utopia seem credible by providing a first-hand account. By showing Hythloday in the map's corner, Holbein anchors the image in the real world: here is the man who was there, pointing to the place he found. The visual authority of cartography is being borrowed to lend credibility to fiction.

POI #5 — The Sailing Ship

Look for… Beneath the island and slightly to one side, find a small sailing vessel rendered in detail — a Renaissance-era ship with sails, masts, and hull. Notice how its shape relates to the form of the island above it.

Art historians have identified this ship as part of the hidden skull composition: the ship's hull and rigging form the jaw and teeth of a skull when the island above is seen as the cranium. The ship in a utopia map is also symbolically charged — it represents the journey to the impossible ideal, the vehicle that can take you to a place that does not exist. It is simultaneously a real vessel and a memento mori, the carrier of the dead to "no place."

POI #6 — The Absence of Scale and Compass

Look for… Look carefully at the overall map and notice what is missing — there is no scale bar, no compass rose, no rhumb lines, no latitude or longitude markings. The island exists in unmeasured, unlocated space.

Every serious cartographic map of the era included scale, orientation, and coordinate references. Their deliberate absence here tells the reader everything: this is a place that cannot be measured, cannot be navigated to, has no fixed position in the world. Holbein has given Utopia the form of a map while denying it the tools of real geography. The absence is the most sophisticated element of the composition.

Web Links

Digital Viewing (Morgan Library & Museum — full woodcut image):
https://www.themorgan.org/exhibitions/online/holbein/utopia

Print-Optimized Reference (Folger Shakespeare Library, open access image):
https://luna.folger.edu/luna/servlet/s/87w69l

Map 4: Carta Marina — 1539 (Olaus Magnus)

The Cartographer

Olaus Magnus (1490–1557) — born Olof Mänsson in Linköping, Sweden — was a Swedish Catholic archbishop, ecclesiastical diplomat, geographer, and historian whose twelve years of exile in Rome and Venice were transformed into two monumental contributions to human knowledge: the Carta Marina of 1539 and its companion volume, Historia de Gentibus Septentrionalibus (A History of the Northern Peoples, 1555). He arrived in Rome on a diplomatic mission for the Swedish Crown in 1523 and never returned, as Sweden's conversion to Lutheranism under King Gustav Vasa made it impossible for a Catholic bishop to reclaim his position. He channeled this displacement into the most ambitious project of his life.

Historical Facts About the Map

- The Carta Marina et Descriptio Septentrionalium Terrarum ("Marine Map and Description of the Northern Lands") was twelve years in the making, printed in Venice in 1539 from nine folio-sized woodblocks, and measured approximately 1.7 meters wide by 1.25 meters tall when assembled — the first large-scale map of any part of the European continent. It was expensive to produce (Magnus borrowed 440 ducats for printing costs) and relatively few copies were made, which is why only two originals survive today: one at the Bavarian State Library in Munich and one at Uppsala University Library in Sweden.
- Unlike the abstract and theological mappae mundi of the Middle Ages, the Carta Marina was built from personal observation, oral accounts, classical texts, and contemporary voyage reports. Magnus had traveled extensively in Scandinavia — as far north as the Arctic settlement limit and as far west as Nidaros (Trondheim), Norway — before completing his theological training in 1517. His first-hand experience of northern geography was unparalleled among cartographers of the age.
- The Carta Marina was the first map of the Nordic countries to provide place names and regional details at a large scale, including the first reasonably accurate depiction of the Baltic Sea and the Gulf of Finland and Gulf of Bothnia. Its coverage extended from northern Germany in the south to the Barents Sea in the north, and from Greenland in the west to Russia in the east.
- The map's sea monsters — over 40 distinct creatures depicted in the Norwegian Sea and Atlantic waters, each labeled with a letter and keyed to descriptions in the margin — were not decorative fancies but represented Magnus's genuine attempt to document marine life based on sailor testimony. Many are exaggerated or invented, but some (the orca, the whale, the walrus, the giant squid) are clearly based on real animals. These images were copied by subsequent cartographers and appeared on maps as late as the early 18th century.
- Magnus later supplemented the map with his 850-page Historia de Gentibus Septentrionalibus (1555), a comprehensive encyclopedia of northern culture, nature, climate, and history in 22 books and 476 chapters. It was eventually translated into Italian, German, English, and Dutch, and for two centuries served as the primary European authority on Scandinavia.

Points of Interest — Scavenger Hunt Style

POI #1 — The Great Sea Serpent (Sea Orm)

Look for… In the Norwegian Sea, west and north of the Norwegian coastline, look for a massive serpentine creature with a dragon-like head — described in the map's key as a serpent 200 feet long — coiling around a helpless sailing ship.

The Sea Orm (or Stoorworm in Scottish tradition) was one of the most feared creatures in northern maritime folklore. Magnus wrote in his Historia that it could raise its head beside a ship and pluck a sailor from the deck. Some scholars believe descriptions of the Sea Orm may have been based on sightings of giant oarfish, giant squid, or rare whale behaviors. This single image had enormous influence: the sea serpent became a fixture on maps of the North Atlantic for 150 years after Magnus drew it here.

POI #2 — The Prister Attacking a Ship

Look for… Look for one or two whale-like creatures labeled with the letter "P" in the margin key — enormous animals shown rising aggressively toward a sailing vessel, with crew throwing barrels overboard to distract them.

The Prister was described by Magnus as an aggressive whale-like monster that sailors distracted by throwing empty barrels for them to play with, while a man on the sterncastle played a loud trumpet (since the creatures allegedly fled from sharp sounds). The barrel-throwing story survives as a genuine sailor's strategy in folk traditions from Lofoten, Norway, to Shetland and Orkney — suggesting it may contain a real kernel of navigational practice, however embellished the monster.

POI #3 — The Beached Whale at the Faroe Islands

Look for… Near the Faroe Islands, look for an illustration of a large whale stranded on a beach, with small human figures busily cutting it up, and what may be a piper or musician nearby.

This scene depicts the culturally important practice of whale butchering, which was a critical food source for isolated island communities in the North Atlantic. The piper is a real detail — music was traditionally played during communal butchering work on Faroese and Orcadian beaches. This image is one of dozens on the Carta Marina that document genuine cultural practices of the northern peoples, giving the map the character of an illustrated ethnography as much as a geography.

POI #4 — Scenes of Daily Life on Land

Look for… On the Scandinavian landmass itself, look for numerous small illustrations of people engaged in everyday activities — reindeer pulling sleds, hunters on snowshoes, knights fighting with cannon, and figures in Nordic dress.

Magnus used the land areas of the Carta Marina as an illustrated encyclopedia of northern life, culture, and history. He included scenes visible only to those who had traveled there: reindeer-drawn sleds in Lapland, snowshoe hunting, the specific church of St. Magnus in Shetland. These were not decorative generic "exotic people" images but specific, observed cultural records, making the Carta Marina the most detailed surviving visual documentation of 16th-century Scandinavian life.

POI #5 — Iceland with Volcanic Activity

Look for… Find the large island labeled Iceland (Islandia) in the upper left of the map, and look for symbols of volcanic eruptions or fire — the smoking or erupting peak of Mount Hekla, shown as an active volcano with flames.

Mount Hekla was a genuine obsession for medieval and Renaissance Europeans, who believed it was literally an entrance to Hell — one of the most widely repeated geographic beliefs of the Middle Ages. Magnus depicted it as actively erupting, which it regularly was, and this reinforced its hellish reputation. For generations of European map readers, Iceland's location on the map was inseparable from its identity as the gateway to the underworld.

POI #6 — The Polar Bear on an Ice Floe

Look for… In the waters north of Iceland or Greenland, look for a white bear depicted on a floating ice sheet in the open Arctic Ocean.

This early cartographic appearance of the polar bear reflects Magnus's access to Norse and Icelandic oral accounts of the far north. Polar bears had been known in medieval Europe — they were occasionally brought south as royal gifts — but their depiction on a geographic map was novel. The image marks the northern limit of recorded human observation: beyond this ice bear, the map becomes speculation.

POI #7 — The Monster Key in the Lower Left Corner

Look for… At the bottom left of the map, find a block of text serving as a key or legend, with letters corresponding to the various sea creatures depicted in the ocean areas. Small illustrations may accompany the text.

This key transforms the Carta Marina from a mere curiosity into a systematic zoological catalog — each monster is named, lettered, and described. This scholarly apparatus reflects Magnus's genuine belief that he was documenting real creatures, and it gave the map academic authority that subsequent cartographers cited and reproduced. The key was the mechanism by which Magnus's sea monster tradition was transmitted to later generations of mapmakers.

Web Links

Digital Viewing (Uppsala University Library — all nine sheets, zoomable): https://www.ub.uu.se/find-and-order/carta-marina/

Print-Optimized Reference (James Ford Bell Library, University of Minnesota, full color facsimile): https://umedia.lib.umn.edu/item/p16022coll184:7550

Map 5: Septentrionalium Terrarum — 1606 (Mercator & Hondius)

The Cartographer

Gerardus Mercator (1512–1594) — born Gerard de Kremer in Rupelmonde, Flanders — is one of the most consequential figures in the entire history of cartography, inventor of the Mercator projection (1569) and the first to use the word "Atlas" for a bound collection of maps. He was also a globe-maker, mathematician, philosopher, and instrument-maker whose 1569 wall map of the world transformed navigational practice for centuries. Jodocus Hondius (1563–1612) was a Flemish engraver and cartographer from Ghent who purchased Mercator's atlas plates after the death of Mercator's son, revised and expanded the collection with 36 additional maps, and republished it in Amsterdam in 1606, launching the most commercially successful map series of the 17th century.

Historical Facts About the Map

- The Septentrionalium Terrarum Descriptio ("Description of the Northern Lands") is the first separate printed map of the Arctic and North Polar region — the world's first map devoted exclusively to the area around the North Pole. Mercator originally conceived it as a small inset in the lower-left corner of his monumental 1569 world map (because the Mercator projection makes polar regions infinitely large and impossible to include in the main map), and his son Rumold first published it as a standalone map in the posthumous third volume of the Mercator Atlas in 1595. The 1606 Hondius edition shown here is the second state of this landmark map.
- In creating the map, Mercator relied on the Inventio Fortunatae — a now-lost medieval account of an Arctic voyage by an English Franciscan monk, possibly from the 14th century — and on letters from the mathematician Jacobus Cnoyen, which described the Arctic as containing a central whirlpool or Magnetic Sea surrounded by four large landmasses. Though completely fictional, this concept dominated Arctic cartography for nearly a century.
- The map shows "El Streto de Anian" (the Strait of Anian) separating North America from Asia at the top-right — a legendary northwest passage route to the Pacific. Its appearance on this authoritative Mercator map helped drive decades of Arctic exploration by English, Dutch, and later Russian explorers desperately seeking a shortcut to the East Asian trade routes. Martin Frobisher, John Davis, Henry Hudson, and Willem Barentsz all went north in part because maps like this one suggested an open passage existed.
- When Jodocus Hondius acquired Mercator's map plates in 1604, he refreshed and updated the engraving, incorporating new information from Dutch Arctic voyages. The 1606 second state depicts Nova Zembla (Novaya Zemlya) as a single island with an incomplete coastline — reflecting the partial surveys completed by Barentsz in 1596–97. This is one of the map's most historically precise details: incomplete knowledge drawn as incomplete coastline, an unusually honest cartographic convention.
- This map also contains one of the first printed uses of "California" to refer to the northern Pacific coast of North America — applied here to Spanish territory in the upper-right area

of the map. The 1606 edition of Mercator's atlas is credited with being the first time this geographical name appears in print outside of its original application to the peninsula of Baja California.

Points of Interest — Scavenger Hunt Style

POI #1 — The Polar Whirlpool and the Four Arctic Islands

Look for… At the very center of the map — directly at the North Pole — look for a circular design representing a magnetic sea or whirlpool, surrounded by four large landmasses arranged symmetrically in the cardinal directions.

Mercator derived this design from the Inventio Fortunatae, an account he believed to be a reliable medieval source. He described it in a letter to the English geographer John Dee: the Arctic contained a polar sea with a great black magnetic rock at its center, drawing all compass needles and swallowing the sea itself into the earth. This idea was taken entirely seriously by educated Europeans — it explained why compasses pointed north. The whirlpool is one of history's most consequential geographic myths.

POI #2 — El Streto de Anian — The Northwest Passage

Look for… In the upper-right area of the map, between the landmass of North America and the coast of Asia, look for a narrow labeled strait — "El Streto de Anian" — marking the presumed water passage between the two continents.

The Strait of Anian was derived from Marco Polo's mention of a Chinese province of Ania on a large gulf. Its appearance on this Mercator map lent enormous authority to the idea of a northwest (and northeast) passage to the Pacific. Expeditions searching for this passage — by Frobisher (1576–78), Davis (1585–87), Hudson (1607–11), and Barentsz (1594–96) — collectively opened the Arctic to European exploration and contributed to the eventual mapping of North America's northern coastline.

POI #3 — Nova Zembla with Incomplete Coastline

Look for… Find the large island labeled Nova Zembla (modern Novaya Zemlya, in Russia) on the right side of the map, between Europe and the Arctic Ocean. Notice that its coastline appears unfinished or dashed in certain sections.

The deliberately incomplete coastline of Nova Zembla reflects actual cartographic honesty: Dutch explorer Willem Barentsz had surveyed the western coast of the island in 1596 before being trapped by ice and dying there in 1597. Hondius incorporated Barentsz's partial survey into the 1606 edition but left the uncharted portions unfinished rather than inventing coastlines — a notably principled cartographic decision that distinguishes this map from many contemporaries.

POI #4 — Greenland and Its Mysterious Interior

Look for… Find the large landmass on the left side of the map labeled Groenland (Greenland). Notice that while its western coast is roughly sketched, its interior is largely blank or filled with uncertain symbols.

Greenland remained one of the most poorly understood landmasses in European geography throughout the 16th and early 17th centuries. Norse settlers had been there since the 10th century but the colonies had died out by the 15th century, and accurate survey information was minimal. Mercator's sparse interior reflects genuine ignorance rather than negligence — the blank interior of Greenland on this map is a cartographic admission of the limits of knowledge.

POI #5 — The Pygmy Island

Look for… In the lower-right area of the map, look for a small island with an incomplete coastline — sometimes labeled with a reference to pygmies or small peoples.

The inclusion of a "pygmy island" reflects the persistent medieval European tradition of placing exotic or monstrous races at the edges of the known world. For Mercator, drawing on the same literary tradition that had populated medieval mappae mundi with dog-headed people and one-eyed giants, the Arctic was the last frontier of geographic imagination. The incomplete coastline of this island in the 1606 edition is one of the cartographic distinguishing features of this specific state.

POI #6 — England and Northern Europe Below the Pole

Look for… In the lower portion of the map, below the polar regions, find the recognizable outlines of Britain, Scandinavia, and northern Russia, giving geographic context to the speculative polar zones above.

By anchoring the speculative polar regions to the accurately surveyed coastlines of northern Europe below, Mercator created a map that blended documented reality with informed imagination. The transition from the accurately drawn coast of Norway to the invented polar continents above was gradual and subtle — a masterclass in making the unknown seem continuous with the known. This technique is what made the map so persuasive and so influential.

Web Links

Digital Viewing (Wikimedia Commons, full resolution 1606 Hondius edition): https://commons.wikimedia.org/wiki/File:1606_Mercator_Hondius_Map_of_the_Arctic_(First_Map_of_the_North_Pole)_-_Geographicus_-_NorthPole-mercator-1606.jpg

Print-Optimized Reference (Geographicus Rare Antique Maps — detailed description with image): https://www.geographicus.com/P/AntiqueMap/NorthPole-mercator-1606

Map 6: Nautical Chart (Mao Kun Map / Zheng He Navigation Chart) — Published 1628

The Cartographer

Zheng He (1371–1433/35) — born Ma He into a Muslim family in Yunnan Province, China — was the greatest admiral in Chinese history and one of the most extraordinary navigators in the history of the world. A court eunuch who rose to become Grand Eunuch and Admiral of the Treasure Fleet under Emperor Yongle of the Ming dynasty, he commanded seven epic voyages across the Indian Ocean between 1405 and 1433, reaching Southeast Asia, India, Persia, Arabia, and the East African coast as far as modern Kenya. The nautical charts attributed to his voyages were compiled by Mao Yuanyi and published posthumously in the military treatise Wubei Zhi in 1628 — nearly 200 years after Zheng He's last voyage.

Historical Facts About the Map

- The Mao Kun Map (full Chinese title: Zheng He Hanghai Tu, or "Zheng He's Navigation Map") is not a single map but a set of navigation charts embedded in the Wubei Zhi (A Treatise on Armament Technology), a Ming dynasty military encyclopaedia compiled by scholar Mao Yuanyi in 1621 and published in 1628. The charts were originally a continuous strip map measuring 20.5 cm × 560 cm that could be rolled up; they were divided into 40 folios for inclusion in the printed book. The charts chart sailing routes from Nanjing, China, south along the Vietnamese coast, through the Straits of Malacca, across the Indian Ocean, around the Arabian Peninsula, and south along the East African coast.
- The map is the earliest known Chinese cartographic document to give an adequate representation of Southern Asia, Persia, Arabia, and the East African coast. Over 300 named places outside China are identified, of which more than 80% can be confidently located today. Place names include Calicut (modern Kozhikode, India), Hormuz (Persia), Aden (Yemen), Mombasa and Mogadishu (East Africa), and dozens of others — representing the commercial and diplomatic network of Zheng He's fleet.
- The map does not use latitude and longitude. Instead, it presents navigational instructions in a practical "sailing directions" format using a 24-point Chinese compass system, sailing times or distances, current information, and stellar altitude observations (50 are recorded). It was a working navigational document for experienced pilots, not a geographic representation for scholars.
- Scholarly analysis suggests the charts were not the work of a single cartographer but were compiled over multiple voyages, with information added and corrected after each expedition. Arabic terms appear for some Indian Ocean features (e.g., Jazirah for island), suggesting that Arab pilots with expertise in Indian Ocean navigation contributed to the charts — a remarkable early example of intercultural geographic knowledge transfer.
- Most records of Zheng He's voyages were deliberately destroyed during later Ming dynasty policy reversals that banned oceanic voyaging. The fact that these charts survived — hidden in the private library of the Mao family before being published in 1628 — makes them among the most precious surviving documents of Chinese maritime

history. Zheng He's seven voyages, which predated Vasco da Gama's round-Africa voyage by decades, involved fleets of up to 300 ships carrying 28,000 men — a scale of maritime enterprise that Europe would not match for another century.

Points of Interest — Scavenger Hunt Style

POI #1 — The Starting Point: Nanjing and the Chinese Coast

Look for… At the beginning of the chart sequence (the rightmost or first section), find the starting point of the navigational route — the Chinese capital of Nanjing and the coast of Fujian Province, with place names written in Chinese characters.

Nanjing was the seat of Emperor Yongle and the departure point for all of Zheng He's treasure voyages. The Jinghai Temple — the "Temple of Tranquil Seas" — near Nanjing was built by Yongle to pray for the safety of the fleet and still exists today. By beginning the chart at Nanjing, the mapmakers were asserting Chinese imperial authority as the organizing principle of the entire Indian Ocean world: all routes begin and end at the center of the Ming Empire.

POI #2 — The Straits of Malacca

Look for… Midway through the chart, find the narrow passage between the Malay Peninsula and Sumatra — a constriction in the sea route where the charts show multiple ports and course headings. Look for the label for Malacca itself.

The Straits of Malacca was the critical chokepoint of Indian Ocean trade, controlling access between the South China Sea and the Indian Ocean. Zheng He's establishment of a permanent Chinese presence at the Sultanate of Malacca in 1403 transformed it into the most important commercial center in Southeast Asia. Chinese diplomatic influence at Malacca was so strong that it became a tributary state of the Ming Empire — and the Strait of Malacca has remained one of the world's busiest shipping lanes ever since.

POI #3 — The Compass Rose Directions and Sailing Instructions

Look for… Throughout the charts, look for clusters of Chinese characters alongside the sea routes — these are not place names but sailing directions, giving compass headings (using a 24-point system) and estimated sailing times between ports.

This practical notation system shows that the Zheng He charts were designed for use at sea, not for display. The 24-point Chinese compass (which predates the European 32-point system) divided the horizon with sufficient precision for open-ocean navigation. The combination of compass heading and sailing time gave pilots the equivalent of a course-and-speed calculation that could be verified against celestial observations — a sophisticated integrated navigation system.

POI #4 — Hormuz and the Arabian Peninsula

Look for… In the western portion of the chart, find the Persian Gulf opening, with the island of Hormuz (labeled in Chinese transliteration) near its mouth, and the Arabian Peninsula beyond.

Hormuz was the great entrepôt of the medieval Indian Ocean — the crossroads of Chinese, Indian, Arab, and Persian trade. Zheng He visited it on multiple voyages, establishing diplomatic relations with the local sultanate. The chart's detail of Hormuz and the Arabian Gulf reflects the extraordinary commercial sophistication of the Ming voyages: this was not exploration for its own sake but state-sponsored commercial and diplomatic engagement with the wealthiest trading ports in the world.

POI #5 — The East African Ports: Mombasa and Mogadishu

Look for… At the far western end of the chart, find the East African coastline labeled with Chinese phonetic transcriptions of African port names — Mombasa (listed as Manba/Manbasa) and Mogadishu (Mujulushu) among others.

Zheng He's voyages reached the Swahili Coast of East Africa on at least three of the seven expeditions (1418, 1421–22, and 1431–33), and the East African section of the chart records the furthest extent of Chinese geographical knowledge. The presence of these African ports on a Chinese naval chart — with accurate relative positions derived from direct navigation — is one of the most remarkable facts in the pre-modern history of cartography. China had sailed to Africa and back more than 70 years before Vasco da Gama rounded the Cape of Good Hope.

POI #6 — Stellar Altitude Observations

Look for… At certain points along the route, look for annotations indicating stellar altitude — numerical notations accompanied by the names of stars used for celestial navigation.

The 50 stellar altitude observations recorded in the Mao Kun Map represent one of the earliest surviving systems of celestial navigation in the world. Chinese navigators measured the altitude of specific stars above the horizon (particularly the Southern Cross and Polaris) to determine their latitude. This technique, independently developed from European celestial navigation, allowed Zheng He's fleet to navigate with confidence across 12,000 kilometers of open ocean — a feat of navigational astronomy that has no equal in the pre-modern world outside of Polynesia.

Web Links

Digital Viewing (UCSB Barbierilow, full interactive chart — archive.org copy):
https://web.archive.org/web/20190628043412/https://barbierilow.faculty.history.ucsb.edu/Research/ZhengHeMapZoomify/ZhengHe.htm

Print-Optimized Reference (Wikipedia — Mao Kun Map overview with full chart image):
https://en.wikipedia.org/wiki/Mao_Kun_map

Map 7: Hereford Mappa Mundi — c. 1300

The Cartographer

Richard of Haldingham and Lafford (also known as Richard de Bello, active c. 1250–1305) was an English clergyman who served as Treasurer of Lincoln Cathedral between 1250 and 1260, held a living at Stoke Talmage in Oxfordshire, and was eventually appointed to a prebendal stall at Hereford Cathedral in 1305. He is known to cartographic history through a single inscription on the map itself — written in Norman French in the lower left-hand corner — which reads: "Let all who have this story, or shall hear or see it, pray to Jesus in God for pity on Richard of Haldingham and Lafford who has made and drawn it, that joy in heaven be granted to him." This prayer-as-signature is one of the most personal statements any medieval cartographer left on his work.

Historical Facts About the Map

- The Hereford Mappa Mundi is the largest surviving medieval map of the world, drawn in pen and ink (with red, gold, blue, and green accents) on a single sheet of calfskin parchment measuring approximately 1.58 x 1.33 meters (about 64 by 52 inches). Dating from around 1300, it was most likely created in Lincoln before being transferred to Hereford Cathedral, where it has remained for more than 700 years and where it is still displayed today in a specially built chamber adjacent to the Cathedral's celebrated Chained Library.
- The map contains approximately 500 separate drawings, including depictions of 420 cities and towns, 15 Biblical events, 8 mythological scenes, 33 plants and animals, 32 illustrations of people from various regions, and dozens of exotic creatures and monstrous races. It draws on sources including Ptolemy's Geographia, the Polychronicon tradition, Isidore of Seville's Etymologiae, the Alexander Romance, the Wonders of the East, and numerous other classical and medieval texts.
- Scholars believe the map originally formed the central panel of a triptych — a three-paneled hinged display — with the angel Gabriel on the left panel and the Virgin Annunciate on the right. This association with the cult of the Virgin Mary was central to the map's function: it may have served as a visual meditation for pilgrims visiting Hereford Cathedral, and recent scholarship strongly links it to the promotion of the cult of Thomas de Cantilupe, a Hereford bishop canonized as a saint in 1320.
- In 1988, the Diocese of Hereford faced a financial crisis severe enough that the Dean and Chapter proposed selling the Mappa Mundi. The public outcry was enormous: major donations from the National Heritage Memorial Fund, J. Paul Getty, and thousands of individual subscribers raised enough money not only to keep the map in Hereford but to build a dedicated new library building (designed by architect Sir William Whitfield) to house it properly. The new Mappa Mundi and Chained Library building opened in 1996.
- Although described by the art historian Jill Tattersall as "without parallel the most important and most celebrated medieval map in any form, the most remarkable illustrated English manuscript of any kind," the map is not a geographic document in the modern sense — it is a visual encyclopedia, a theological statement, and a work of art that happens to use the conventions of cartography to organize human knowledge. Its

purpose was not to help travelers find their way but to help Christians understand their place in the divine order of Creation.

Points of Interest — Scavenger Hunt Style

POI #1 — Christ in Majesty at the Top

Look for… At the very top (eastern edge) of the circular map, look for a large enthroned figure of Jesus Christ, depicted in glory — possibly holding orbs, with angels and the blessed surrounding him. Below this scene the map begins.

The map is oriented East-up, and the reason is theological: East is the direction of paradise, the rising sun, and the Last Judgment. Christ presiding over the eastern summit is simultaneously the sun rising over creation and the judge who will receive the souls at the end of time. The Hereford map reads from top to bottom as a narrative of salvation history — from the garden of Paradise at the top to the world of human sin and suffering below, with Jerusalem at the center as the axis of redemption.

POI #2 — Jerusalem at the Exact Center

Look for… Find the city of Jerusalem positioned at the geometric center of the circular map. It is usually depicted as a circular walled city with a prominent symbol of the Crucifixion or the Church of the Holy Sepulchre at its heart.

Jerusalem's placement at the center is a theological statement of absolute priority: this is the most important place on earth, the site of Christ's death and resurrection, the navel (omphalos) of the Christian world. Its central position was copied from the Roman tradition of placing Rome at the world's center — the Hereford map replaced Rome with the Holy City, asserting that the new axis of the world was not imperial but divine. For medieval viewers, this was not a distortion of geography but an entirely correct hierarchy of spiritual importance.

POI #3 — The Garden of Eden — Paradise

Look for… At the very top of the map, just below or near the figure of Christ, find a walled garden or enclosed space with the figures of Adam and Eve, a serpent, and (often) the four rivers of Paradise flowing outward.

In medieval geography, Paradise was a real place — physically located in the far East, the source of the four great rivers of the world (Nile, Tigris, Euphrates, Ganges). The Hereford map places it at the literal top of the world — the most elevated, most sacred point. The inclusion of Adam and Eve being expelled from Eden at the dawn of the map is the opening scene of human history: the map begins with Creation and expulsion, moves through the biblical and classical world, and ends at the Last Judgment in the top corner — a complete visual narrative of the Christian story.

POI #4 — Monstrous Races at the Edges of the World

Look for… Around the outer rim of the map, particularly in Africa and the far East, look for drawings of fantastical beings — among them a man using his own enormous foot as a sunshade (the Sciapod), dog-headed people (Cynocephali), giants, headless people with faces on their chests (Blemmyae), and other strange figures.

These "monstrous races" derive from classical sources — principally Pliny the Elder's Natural History and the Wonders of the East — and were a standard feature of medieval world maps. Their placement at the margins reflects the medieval understanding of geography as a spectrum from the sacred center (Jerusalem) outward to the savage and the monstrous edge. They were

not ignorance but scholarship: the map was citing respected classical authorities. The margins of the known world were the habitation of those who had not yet received Christian civilization.

POI #5 — The Red Sea Rendered in Red

Look for… In the right-hand portion of the map (the southern region, representing Africa and the Indian Ocean area), look for a body of water rendered in red ink — the Red Sea, colored to match its name.

The Red Sea's red color is one of the most charming literalisms of the Hereford map. The mapmaker knew that the Red Sea was called red, and so he painted it red — a perfect example of medieval cartography's tendency to make names visually literal. It also demonstrates the mapmaker's access to actual geographic knowledge: the Red Sea appears in roughly the correct position relative to Africa and the Arabian Peninsula, drawn from classical Roman geographic texts.

POI #6 — The Inscription of Richard of Haldingham

Look for… In the lower-left corner of the map, look for a block of text written in Norman French — the prayer and attribution of the mapmaker, asking for prayers in exchange for his work.

This inscription is one of the most personal documents in the history of cartography. Richard did not sign the map with his name as an assertion of authorship in the modern sense — he wrote a prayer, asking future viewers to intercede for his soul in return for the gift of the map. The map was an act of devotion as much as an act of scholarship. In writing "pray for Richard of Haldingham and Lafford who made and drew this," he encoded his own immortality into one of the greatest works of medieval art.

POI #7 — The Roman Emperor Augustus

Look for… In the lower-left area of the map, near the western edge, look for a figure identified as the Roman Emperor Augustus, depicted giving orders or presiding over the world. Latin text nearby may reference his commission of a world survey.

Augustus's presence on the map connects Christian medieval geography to the classical Roman tradition. According to the map's sources, Augustus ordered a survey of the entire world by his son-in-law Agrippa, producing a lost world map that became the foundation of much medieval geographic knowledge. By including Augustus on the map, Richard connected the origins of geographic knowledge to imperial Roman authority — and then to the Christian truth that superseded it. The map traces a lineage from Roman empire to Christian empire, with Jerusalem replacing Rome at the center of the world.

Web Links

Digital Viewing (Hereford Cathedral — official high-resolution interactive map): https://www.themappamundi.co.uk/

Print-Optimized Reference (Factum Foundation — 3D scan documentation and high-resolution image): https://factumfoundation.org/our-projects/digitisation/the-hereford-mappa-mundi/

Notes on Using These Maps in ArcGIS Story Maps

Each Point of Interest entry in this guide has been formatted for direct use as a callout panel in an ArcGIS Story Map. Use the POI title as your panel header, the "Look for…" line as your

visual prompt for viewers, and the following paragraph as your body text. The digital links provide high-resolution zoomable images suitable for interactive panels, while the print-optimized links are appropriate for preparing large-format printed displays or exhibition materials.

For the best Story Map experience, consider pairing each POI callout with a cropped or zoomed screenshot of the relevant area of the map. All seven maps are either in the public domain or freely available through institutional repositories, and most of the linked libraries explicitly permit educational and non-commercial use of their digital images.